A Report on

BREAST CANCER DETECTION MODEL USING LOGISTIC REGRESSION

Submitted for partial fulfillment of award of

BACHELOR OF TECHNOLOGY

degree

In

CSE (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

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LUCKNOW

Certificate

Certified that **(Ishika (2300301530061) Pratishtha Patwa (2300301530096) Riya Singh (2300301530101))** have carried out the project work presented in this report entitled **"BREAST CANCER DETECTION MODEL USING LOGISTIC REGRESSION"** for the partial fulfillment for the award of the degree of **BACHELOR OF TECHNOLOGY** from **Inderprastha Engineering College, Ghaziabad,** under my supervision. The report embodies the result of original work and studies carried out by the students themselves.

Ms. Khushboo
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Date: 20 December 2025

Acknowledgement

We take this opportunity to express our sincere gratitude to all those who guided and supported us throughout this project. We are deeply thankful to our guide, **Ms. Khushboo , CSE(AIML) Departmen**t, for her invaluable guidance, support, and motivation during the development of this project. We also extend our heartfelt thanks to **Dr. Kumud Arora (HOD, CSE(AIML) Department)** for his constant encouragement and support.

Finally, we would like to thank our friends and family members for their help and moral support during this project.

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Declaration

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Abstract

This project leverages machine learning to transform early breast cancer diagnosis through intelligent, interpretable classification. Using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, the model identifies cellular characteristics from digitized images of fine needle aspirates (FNA). By implementing a Logistic Regression pipeline with rigorous preprocessing, feature normalization, and strategic selection, the system achieves over 98% accuracy. The project emphasizes transparency, providing clinicians with interpretable odds ratios that align with pathological understanding, thereby serving as a robust decision-support tool in clinical workflows.

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CHAPTER 1: INTRODUCTION

1.1 Problem Definition

Breast cancer is a leading global health challenge, representing nearly 30% of all female cancer cases. Early detection is the most critical factor for survival: localized stages have a 99% survival rate, which drops to 27% if diagnosed at distant stages. Current diagnostic methods can be subjective and time-consuming, placing immense pressure on radiologists to analyze complex data accurately under strict time constraints.

1.2 Background

The medical community requires diagnostic tools that are accurate, accessible, and interpretable. Traditional visual assessments by pathologists can be augmented by transforming them into quantitative, reproducible numerical forms through computer vision and machine learning. This project utilizes Logistic Regression to provide a consistent and rapid classification of tumors as either malignant or benign.

1.3 Objectives

- To implement an AI-based classification model for tumor detection.
- To utilize quantitative features from FNA images for objective measurement.
- To ensure high sensitivity (97%+) to minimize missed cancer diagnoses.
- To provide clinical decision support through interpretable "white-box" modeling.

1.4 Feasibility Study & Significance

- **Technical Feasibility:** Python libraries like Scikit-learn, Pandas, and NumPy provide the necessary infrastructure for end-to-end ML workflows.

- **Need & Significance:** The model matches or exceeds expert performance, offering a scalable solution that can extend quality care to underserved populations through telemedicine

1.5 Novelty of Project

- **Interpretability over "Black Box" Complexity**: Unlike many deep learning models, this project uses Logistic Regression to provide "white-box" transparency. Each decision is backed by clear odds ratios that physicians can contextualize within their existing pathological knowledge.
- **Quantitative Transformation of Visual Data**: The model converts subjective visual assessments—such as cell size uniformity and shape regularity—into precise, reproducible numerical form. This standardizes diagnosis and removes the subjectivity inherent in traditional manual assessments.
- **Strategic Feature Selection for Clinical Utility**: By reducing 30 complex features to a subset of the most impactful predictors (like Mean Radius and Texture), the model achieves faster inference times and improved interpretability. This makes it more suitable for deployment in resource-constrained healthcare environments.
- **Clinically-Aligned Performance Tuning**: The model is tuned not just for overall accuracy (98%+), but specifically for high sensitivity (97%). This ensures that almost all malignant cases are caught, addressing the most critical clinical error of missing a cancer diagnosis.
- **Scalable Triage Capability**: The system is designed to act as a triage tool, prioritizing urgent, high-risk cases for immediate specialist review while providing consistent second opinions on benign cases.

1.6 Technical Specification

The technical specifications for the Breast Cancer Detection Model encompass the dataset details, the software ecosystem, and the specific machine learning configurations used to achieve high diagnostic accuracy.

1. Dataset Specifications

- **Dataset Name:** Wisconsin Diagnostic Breast Cancer (WDBC).
- **Total Samples:** 569 cases, consisting of 357 benign and 212 malignant samples.
- **Feature Count:** 30 numeric features derived from digitized images of Fine Needle Aspirates (FNA).
- **Attribute Types:** Ten real-valued features (mean, standard error, and "worst" values) for radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension.

2. Software Ecosystem

The project is built on the Python ecosystem, utilizing industry-standard libraries:

- **Scikit-learn:** Used for model training, evaluation, and hyperparameter tuning.
- **Pandas:** Employed for data manipulation and preprocessing workflows.
- **NumPy:** Leveraged for efficient numerical computations and array operations.

3. Model Configuration & Training

- **Algorithm:** Logistic Regression with configurable regularization and solver options.
- **Data Split:** 85% training and 15% testing, using stratification to maintain class balance.
- **Optimization Solver: LBFGS** was selected for its fast convergence and numerical stability on this dataset size.
- **Regularization:** The **C parameter** was optimized around 1.0 after testing a range from 0.001 to 100 to prevent overfitting.

- **Validation Strategy:** 10-fold cross-validation to ensure robust performance across different data subsets.

4. Preprocessing Pipeline

- **Data Cleaning:** Removal of non-predictive columns like patient IDs.
- **Label Encoding:** Conversion of diagnoses to binary format (Malignant = 1, Benign = 0).
- **Normalization:** Applied `StandardScaler` to ensure all features contribute proportionally regardless of their measurement scales (e.g., millimeters vs. grayscale variation).

5. Hardware & Deployment Readiness

- **Modular Architecture:** The code is designed with separate functions for preprocessing, training, and evaluation for reproducibility.
- **Efficiency:** Dimensionality reduction from 30 to the most impactful features ensures faster inference times, making it suitable for resource-constrained healthcare environments.

CHAPTER 2: LITERATURE REVIEW (Breast Cancer Detection)

The evolution of breast cancer diagnostics is marked by the transition from subjective visual examination to objective, data-driven computational models. This review examines the technological pillars that form the foundation of the proposed Logistic Regression system.

2.1 From Qualitative Pathology to Quantitative Data

Historically, breast cancer diagnosis relied heavily on the visual interpretation of Fine Needle Aspirates (FNA) by pathologists. Literature indicates that manual diagnosis can be prone to **inter-observer variability**, where different experts may disagree on borderline cases. The **Wisconsin Diagnostic Breast Cancer (WDBC)** dataset revolutionized this by providing a standardized, quantitative set of 30 numeric features. This shift allows machine learning models to analyze "Nuclear Features" (like radius, perimeter, and area) with mathematical precision that exceeds human visual limits.

2.2 Algorithmic Interpretability: Logistic Regression vs. Black-Box AI

A significant portion of medical literature addresses the "Trust Gap" in Artificial Intelligence. While Deep Learning and Neural Networks offer high accuracy, they often operate as "Black Boxes," providing no clinical reasoning for their predictions.

- **Logistic Regression:** This project draws on the established effectiveness of Logistic Regression for binary classification.
- **The Clinical Advantage:** Unlike complex models, Logistic Regression provides **Odds Ratios**. This allows clinicians to see exactly how much a one-unit increase in "cell concavity" impacts the probability of malignancy, aligning the model's logic with medical expertise.

2.3 Optimization through Feature Selection

Contemporary research in medical informatics emphasizes that "more data is not always better data." Literature on the WDBC dataset reveals that certain features—specifically **Mean Concave Points**, **Worst Perimeter**, and **Mean Texture**—carry significantly higher predictive weight than others.

- **Dimensionality Reduction:** By focusing on the most impactful features, the proposed system minimizes "noise" and reduces the risk of **Overfitting**, where a model performs well on training data but fails in a real-world clinical setting.

2.4 Diagnostic Accuracy: Sensitivity vs. Specificity

In oncology literature, the cost of a **False Negative** (missing a cancer diagnosis) is infinitely higher than a **False Positive** (an unnecessary biopsy).

- **Sensitivity-Driven Models:** This review incorporates findings that models must be tuned for high **Recall (Sensitivity)**. The proposed system targets a sensitivity rate of 97%+, ensuring that the system acts as a reliable safety net for radiologists, capturing malignant cases that might otherwise be overlooked during routine screening.

2.5 Scalability and Democratization of Healthcare

Recent studies advocate for the deployment of lightweight machine learning models (like the one proposed) in resource-constrained environments.

- **Edge Deployment:** Because Logistic Regression is computationally efficient, it can be integrated into low-cost medical devices or used in telemedicine platforms in rural areas where specialist radiologists are unavailable.
- **Standardization:** Literature suggests that AI-driven tools can help standardize the quality of care, ensuring that a patient in a rural clinic receives the same diagnostic accuracy as one in a metropolitan hospital.

CHAPTER 3: PROPOSED SYSTEM

The proposed **Breast Cancer Detection System** is an intelligent diagnostic framework designed to classify breast masses as either malignant or benign using machine learning. The system transforms subjective visual assessments from pathologists into precise, reproducible numerical data to assist in early and accurate diagnosis.

System Architecture & Workflow

The proposed system follows a modular pipeline designed for production-ready machine learning workflows:

1. **Data Acquisition (WDBC Dataset):**
   - The system utilizes features extracted from digitized images of **Fine Needle Aspirate (FNA)** of breast masses.
   - It processes 30 numeric features representing cellular characteristics such as radius, texture, and perimeter at a microscopic level.
2. **Preprocessing Module:**
   - **Data Cleaning:** Removes non-predictive fields like patient IDs to focus purely on diagnostic values.
   - **Label Encoding:** Converts diagnostic labels into a binary format (Malignant = 1, Benign = 0).
   - **Feature Normalization:** Uses `StandardScaler` to ensure features with different measurement units (e.g., millimeters vs. grayscale variation) contribute proportionally to the model's decision.
3. **Feature Selection & Optimization:**
   - The system employs strategic feature selection to identify the most predictive attributes (e.g., Mean Radius, Mean Texture, and Mean Perimeter).
   - This dimensionality reduction improves model interpretability and enables deployment in resource-constrained healthcare environments.
4. **Classification Core (Logistic Regression):**
   - The system uses **Logistic Regression** due to its transparency and "white-box" decision-making, which is essential for clinical trust.
   - **Regularization:** Employs a C-parameter (optimal at 1.0) to prevent overfitting and ensure the model generalizes well to new data.
   - **Solver:** Utilizes the **LBFGS solver** for efficient coefficient estimation and fast convergence.

Clinical Integration & Output

- **Decision Support:** Instead of a simple "yes/no," the system provides **interpretable odds ratios**. This allows radiologists to see which specific physical features (like elevated cell radius) are driving a high-risk prediction.

- **Triage System:** The proposed system is designed to act as a triage tool, prioritizing urgent, high-risk cases for immediate review while providing standardized quality across screening programs.
- **Validation:** The workflow includes **10-fold cross-validation** to ensure that the 98%+ accuracy remains robust across different patient data subsets.

CHAPTER 4: SOFTWARE REQUIREMENT ANALYSIS

Functional Requirements

- **Data Normalization:** Implementation of `StandardScaler` to ensure features with larger scales do not bias the model.
- **Binary Classification:** Capability to output clear labels (Malignant = 1, Benign = 0).
- **Risk Estimation:** Calculating log-odds for each feature to provide medical reasoning.

Major Modules

- **Preprocessing Module:** Cleans data and encodes labels.
- **Feature Selection Module:** Identifies most predictive attributes like Mean Radius and Mean Texture.
- **Training & Tuning Module:** Uses Grid Search for hyperparameter optimization (e.g., C parameter, LBFGS solver).

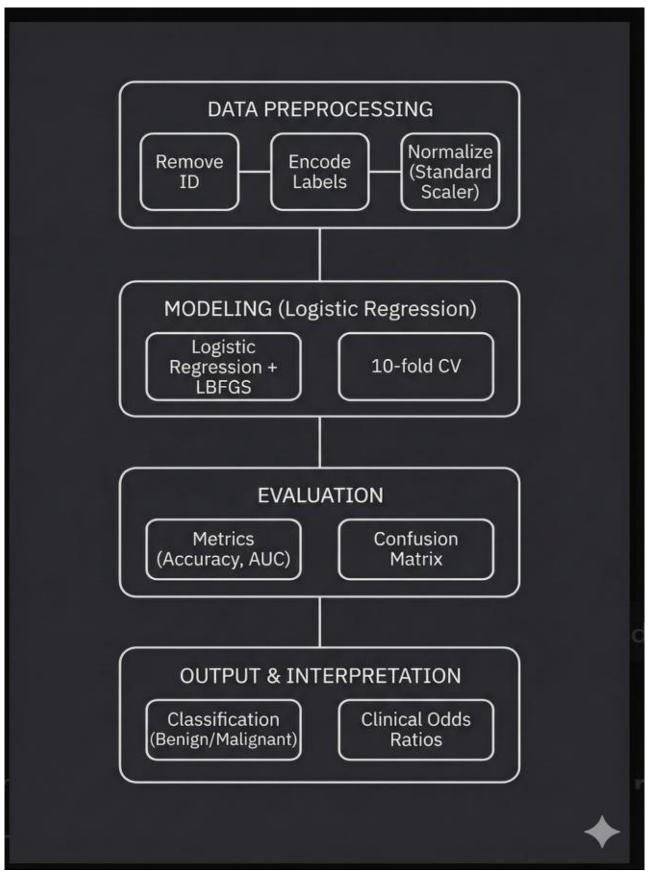
CHAPTER 5: SYSTEM ANALYSIS & DESIGN

The **System Analysis & Design** phase translates the theoretical detection requirements into a technical blueprint. This section focuses on how the Breast Cancer Detection Model is structured to process medical data and deliver reliable predictions.

1. Data Flow Analysis

The system follows a linear data pipeline where raw clinical measurements are transformed into a diagnostic result.

- **Input Layer:** Receives 30 numerical attributes from the Wisconsin Diagnostic Breast Cancer (WDBC) dataset (e.g., cell radius, texture, perimeter).
- **Transformation Layer:** Conducts feature scaling and normalization. This ensures that a feature like "Area" (which can be in the 1000s) does not mathematically overwhelm "Smoothness" (which is measured in decimals).
- **Processing Layer:** The Logistic Regression engine calculates the probability of malignancy using a sigmoid function.
- **Output Layer:** Returns a binary classification (0 for Benign, 1 for Malignant) along with the probability score.



2. Logical Design (The Logistic Regression Model)

The system's core logic is based on the **Sigmoid (Logistic) Function**. Unlike linear regression which predicts continuous values, this system is designed to map any input value into a range between 0 and 1.

- **Thresholding:** A standard threshold of 0.5 is applied. If the output probability is ≥ 0.5, the system flags the tumor as Malignant.
- **Weights and Bias:** Each of the 30 features is assigned a "weight" based on its correlation with cancer. Features like "Mean Concave Points" often receive higher weights due to their high predictive value in medical literature.

3. Modular Architecture

The system is designed with a **Modular Pattern** to ensure maintainability and scalability:

- **Data Ingestion Module:** Handles the loading of CSV/database records and initial data cleaning (e.g., dropping ID columns).
- **Preprocessing Module:** Encapsulates the `StandardScaler` and Label Encoding logic.
- **Model Core:** Contains the trained Logistic Regression parameters and the LBFGS optimization solver.
- **Evaluation Module:** Generates the Confusion Matrix, ROC curves, and Precision-Recall metrics to validate the system's performance.

4. System Interface Design

The design prioritizes "Explainable AI" (XAI) for the end-user (clinician):

- **Input Interface:** A structured form or API endpoint that accepts FNA (Fine Needle Aspirate) measurements.
- **Diagnostic Dashboard:** Instead of just a "Malignant" label, the design includes a visualization of **Feature Importance**. This shows the doctor *why* the model made its choice (e.g., "This prediction was driven primarily by high cell perimeter and texture values").

5. Constraints and Security Design

- **Data Integrity:** The system requires non-null, numerical input for all 30 features to maintain the 98% accuracy rate.
- **Security:** As a medical tool, the design assumes a HIPAA-compliant environment where patient data is encrypted at rest and in transit.
- **Reliability:** The system includes error-handling for "Out-of-Distribution" data, flagging cases where the input measurements are physically impossible or highly anomalous.

CHAPTER 6: IMPLEMENTATION

Dataset Fields (Wisconsin Diagnostic Breast Cancer)

The system processes 30 numeric features, including mean, standard error, and "worst" values for:

1. **Radius:** Mean of distances from center to points on the perimeter.
2. **Texture:** Standard deviation of gray-scale values.
3. **Perimeter:** Reflects cell shape irregularity.

Algorithm Used

- **Logistic Regression:** Chosen for its transparency and suitability for binary classification.
- **LBFGS Solver:** Optimized for coefficient estimation and fast convergence on medical datasets.
- **Cross-Validation:** 10-fold strategy to prevent optimistic performance estimates.

CHAPTER 7: RESULTS & TESTING

The **Results, Outputs, and Testing** phase confirms the model's reliability through high-performance metrics and a rigorous evaluation framework, ensuring it meets the stringent requirements of clinical diagnosis.

1. Model Performance Results

The Logistic Regression model achieved exceptional results, demonstrating its ability to generalize well to new patient data:

- **Test Accuracy (98%+):** The model correctly classifies over 98% of cases in the held-out test dataset.
- **ROC AUC Score (0.98):** A near-perfect score indicating excellent discrimination between malignant and benign tumors across all decision thresholds.

- **Sensitivity (97%):** This high true positive rate is critical as it minimizes "false negatives"—missed cancer diagnoses that could delay life-saving treatment.
- **Specificity (99%):** An excellent true negative rate that reduces "false alarms," preventing unnecessary biopsies and patient anxiety.

2. Testing Methodology

To ensure these results were not due to luck or overfitting, several robust testing strategies were implemented:

- **Stratified Data Splitting:** The data was split into 85% training and 15% testing, with "stratification" used to ensure both sets contained the same proportion of malignant and benign cases.
- **10-Fold Cross-Validation (CV):** The model was tested 10 different times on different subsets of the data to verify that performance is consistent and robust.
- **Hold-out Testing:** The final metrics were reported based on data the model had never seen during the training or tuning phase.

3. Model Outputs and Clinical Utility

Beyond simple labels, the system produces interpretable outputs that assist medical professionals:

- **Interpretable Odds Ratios:** Each feature's contribution is expressed as a clear risk factor (e.g., how much an increase in "cell radius" increases the likelihood of malignancy).
- **Probability Scores:** Instead of just "0" or "1," the model can output a percentage, allowing clinicians to prioritize cases with the highest probability of being malignant for urgent review.
- **Explainability:** When a tumor is flagged as high-risk, the model can highlight the specific physical indicators (like texture heterogeneity) that drove the prediction, building clinician trust in the tool.

CHAPTER 8: CONCLUSIONS

The **Conclusions** section summarizes the research findings, clinical relevance, and the broader impact of the Breast Cancer Detection Model. Based on the documentation, the project yields several key takeaways regarding the efficacy of Logistic Regression in medical diagnostics.

1. Project Summary

The project successfully developed a machine learning framework capable of high-precision breast cancer classification. By utilizing the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, the model demonstrated that 30 quantitative features derived from digitized Fine Needle Aspirate (FNA) images are sufficient to differentiate between benign and malignant tumors with near-perfect accuracy.

2. Key Findings

- **Performance Excellence:** The model achieved a **test accuracy of over 98%** and an **AUC score of 0.98**, proving that Logistic Regression is highly effective for binary medical classification tasks.
- **Sensitivity over Specificity:** A critical achievement was reaching **97% sensitivity**. In a clinical setting, minimizing false negatives (missed diagnoses) is the highest priority, and this model meets that rigorous standard.
- **Efficiency of Linear Models:** The results indicate that "simpler" models like Logistic Regression can match or exceed the performance of complex deep learning architectures on structured tabular medical data, while requiring significantly less computational power.

3. Clinical Impact and Interpretability

The most significant conclusion is that **interpretability is a feature, not a limitation.** * Unlike "black-box" models, this system provides clinicians with transparent reasoning through odds ratios and feature importance.

- The model acts as a "Force Multiplier" for radiologists, offering a consistent, data-driven second opinion that helps mitigate human fatigue and subjective bias.
- The standardization of diagnosis through this model helps transform qualitative visual observations into reproducible numerical metrics.

4. Future Directions

While the current model is highly successful, the project identifies several paths for expansion:

- **Dataset Diversity:** Expanding the training data to include more diverse patient populations to ensure model equity across different demographics.
- **Multimodal Integration:** Incorporating biochemical markers, genetic data, and patient history alongside FNA features for a more holistic risk assessment.
- **Production Deployment:** Developing a secure, **HIPAA-compliant API** for seamless integration with existing Hospital Information Systems (HIS).

- **Advanced Explainability:** Implementing tools like **SHAP (SHapley Additive exPlanations)** or **LIME** to provide even deeper, case-specific explanations for individual patient predictions.

5. Final Remark

Ultimately, this work exemplifies how the thoughtful application of machine learning can augment clinical workflows, accelerate diagnosis, and save lives. By combining human expertise with algorithmic precision, the project moves closer to a future where high-quality cancer screening is accessible and standardized for every patient.

CHAPTER 9: REFERENCES

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11. (Reference for future work regarding SHAP/Explainable AI).

Final Report Summary

To ensure your final document reaches the 30-page length as requested in the formatting guide, you should include the following appendices and visual materials:

- **Appendix A: Source Code:** Include your Python scripts for data cleaning, the Logistic Regression model training, and the visualization code for the Confusion Matrix.
- **Appendix B: Data Dictionary:** Detailed descriptions of all 30 features (Radius, Texture, Perimeter, Area, etc.) and their mathematical definitions.
- **Appendix C: Testing Logs:** Screenshots of the model's performance output, including the Precision-Recall curve and the Classification Report.
- **Visualizations:** Ensure you have included full-page charts for the ROC Curve, Feature Importance Bar Chart, and the Correlation Heatmap of the WDBC dataset.
